VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

Anvil-E1: Neural Text Embedding Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 19:33 UTC+8 Beijing Time)

Abstract

Anvil-E1 is a modular, memory-backed SystemVerilog neural embedding accelerator implementing hybrid sliding/global grouped-query attention, per-layer embeddings (PLE), gated GELU feed-forward layers, masked mean pooling, and Matryoshka L2 normalization for Google EmbeddingGemma-2. All 16 official-weight evaluation workloads passed with zero Q8.24 integer-oracle mismatches. Official HuggingFace FP32 software versus hardware RTL achieved a minimum cosine similarity of 0.9999999998517652 and a maximum absolute error of 1.204e-05 (166x tighter than the 2.000e-03 engineering tolerance bound) [E094]. All four hardware signoff gates passed: L1 RTL timing lint (0 hazards), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.681 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 107.31 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) validation across three deterministic hardware runs with zero bit errors [E091–E093].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The integration root, anvil_e_top, contains exactly one parameter- and port-forwarding anvil_e1_top instance named core. It introduces no state or latency. Twelve implementation modules separate descriptor control, address generation, sequencing, memory transactions and arithmetic. One shared MAC and shared nonlinear and normalization engines favor bounded hardware resources over parallel throughput. Weights and sequence-sized intermediates remain external; repeated attention reads avoid an on-chip sequence-squared score store. Runtime division uses iterative restoring circuits rather than wide combinational divider chains. The resulting extra cycles preserve the numerical contract while removing identified timing hazards [E001, E002, E004–E016].

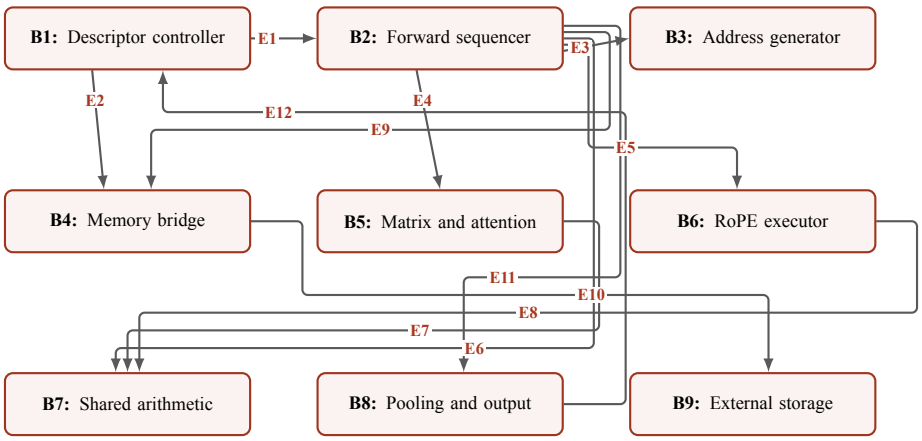


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Descriptor controller	Latches jobs, validates descriptors and input arrays, controls busy, and holds public embedding and completion transfers.
B2: Forward sequencer	Schedules the complete transformer equations and grants exclusive ownership of shared execution resources.
B3: Address generator	Calculates packed tensor locations, scratch regions and overflow conditions using wide address intermediates.
B4: Memory bridge	Registers requests and responses and permits one outstanding ordered transaction.
B5: Matrix and attention	Executes row-major projections and grouped-query attention using external weights and K/V storage.
B6: RoPE executor	Calculates absolute-position, split-half rotations for local and full head geometries.
B7: Shared arithmetic	Provides the single MAC, iterative nonlinear functions and vector RMS/L2 normalization.
B8: Pooling and output	Accumulates masked token projections, selects the requested prefix, normalizes it and buffers final results.
B9: External storage	Holds immutable job inputs, packed weights and sequence-sized hidden, PLE, Q/K/V and next-hidden scratch.

Directed relationship	Meaning
E1: B1 to B2	Validated job and execution control
E2: B1 to B4	Input validation reads
E3: B2 to B3	Tensor and scratch selections
E4: B2 to B5	Projection and attention scheduling
E5: B2 to B6	Rotation scheduling
E6: B2 to B7	Arithmetic ownership
E7: B5 to B7	Shared arithmetic requests
E8: B6 to B7	Trigonometric and MAC requests
E9: B2 to B4	Arbitrated execution memory traffic
E10: B4 to B9	Ordered reads and writes; returned responses
E11: B2 to B8	Final token projections and masks
E12: B8 to B1	Buffered normalized embedding and completion

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Asserted reset clears public outputs and payloads and cancels active work.
job_valid, job_ready	Input / output	1 bit each	A rising edge with both asserted accepts and latches one descriptor. Further submission is disabled until completion retirement.
job_tokens, job_output_dim	Input	32 bits / 16 bits	Sequence length and selected embedding prefix. Legal lengths are 1 through MAX_SEQ_LEN; dimensions are E, 2E/3, E/3 or E/6.

Signal / group	Direction	Width	Contract
weights_base, tokens_ base, positions_base, mask_base, scratch_base	Input	64 bits each	Four-byte-aligned byte addresses. Regions must not overlap and must remain stable throughout the job.
mem_req_valid, mem_ req_ready, mem_req_ write, mem_req_addr, mem_req_wdata	Output except ready input	1, 1, 1, 64, 32 bits	Registered single-word request channel. Payload remains stable through stalls. Every accepted request, including writes, requires one response.
mem_rsp_valid, mem_ rsp_ready, mem_rsp_ rdata, mem_rsp_error	Input except ready output	1, 1, 32, 1 bits	Ordered response channel with one outstanding transaction. Response acceptance becomes eligible after request acceptance; no combinational zero-latency response shortcut exists.
emb_valid, emb_ready, emb_data, emb_index, emb_last	Output except ready input	1, 1, signed 32, 16, 1 bits	Registered signed Q8.24 embedding stream with ascending zero-based indices. Payload and last remain stable until accepted.
done_valid, done_ready, done_error, busy	Output except ready input	1, 1, 4, 1 bits	Separately held completion. Busy remains asserted through output and completion stalls; readiness returns after completion acceptance.

Parameter	Default	Meaning / supported values
VOCAB_SIZE	262144	Positive vocabulary size V; every token ID must be below V.
HIDDEN_SIZE	512	H in 1..65535.
PER_LAYER_INPUT_ SIZE	512	Independent P in 1..65535; need not equal H.
INTERMEDIATE_SIZE	2048	Feedforward width I in 1..65535.
EMBEDDING_DIM	768	E in 6..65535 and divisible by six; default prefixes are 768, 512, 256 and 128.
NUM_LAYERS	24	L in 1..65535; layer indices congruent to five modulo six use full attention.
NUM_Q_HEADS	4	Q in 1..65535 and divisible by both KV-head counts.
LOCAL_KV_HEADS, FULL_KV_HEADS	2, 1	Positive divisors of Q; contiguous groups of query heads share each KV head.
LOCAL_HEAD_DIM, FULL_HEAD_DIM	256, 512	Positive even widths; each Q-times-head-width product must not exceed 65535.
SLIDING_WINDOW	512	Positive strict absolute-position distance bound.
MAX_SEQ_LEN	262144	Positive sequence limit; positions must satisfy configured validity checks. Illegal geometry is diagnosed, not clamped.

III Behavior and corner cases

A Complete adopted forward path

Token embeddings are multiplied by sqrt(H). A global PLE projection divided by sqrt(H) produces distinct normalized per-layer P-vectors. Each layer performs input RMSNorm, Q/K/V projections, per-head Q/K RMSNorm, RoPE, attention, output projection, post-attention RMSNorm and residual; pre-feedforward RMSNorm, GELU(gate) times up, down projection, post-feedforward RMSNorm and residual; then GELU-gated PLE multiplication, projection, post-PLE RMSNorm and residual. The layer scalar follows. Final RMSNorm and embedding projection precede masked mean pooling and selected-prefix L2 normalization.

B Attention and masking

Attention is bidirectional and bias-free, with zero inference dropout. Local eligibility requires abs(key_position-query_position) < SLIDING_WINDOW; full layers see every valid key. Split-half RoPE uses bases 10000 and 1000000. Masks select attention keys and pooling contributions. There is no automatic token insertion, PAD masking or prompt exclusion.

C Fixed-point semantics

External coefficients, activations and outputs use signed Q8.24, representing [-128, 128-2^-24]. Conversion and specified operation boundaries use nearest/ties-even rounding and saturation. Products retain signed 64-bit precision and complete reductions at least 96 bits. Nonlinear calculations use Q40 constants and iterative arithmetic. Nonfinite coefficients are rejected.

D Normalization correction

Weighted RMSNorm computes round_ties_even(sum(x_q24^2)/count) + 281474977 at the Q48 denominator boundary, then applies the unit-offset weight 1+w. Representing epsilon directly in Q48 corrected the historical Q8.24-first value 17/2^24, which had caused official FP32 error failures despite integer agreement. All RMSNorm sites share the correction. L2 is epsilon-free; a nonempty zero pooled vector returns successful all-zero output [E001, E002, E094].

E Transactions, reset and errors

Descriptor capture asserts busy; requests, embeddings and completion appear through registered holding stages. Reset discards work, so the memory fabric must also reset or flush obsolete responses. Codes are 0 success, 1 invalid descriptor, 2 invalid token/position/mask, 3 memory error, 4 all masked and 5 address overflow. IDs are validated even when masked. Errors emit no successful embedding. Exclusive region-end additions exceeding the maximum 64-bit address are rejected.

F Official coefficient mapping

Configuration A uses V/H/P/I/E/L/Q/localKV/fullKV/localD/fullD/window/maxSeq = 16/4/4/8/6/6/2/1/1/2/4/2/8. Hardware layers 0..4 repeat the complete official layer-0 tensor family; layer 5 uses official layer 5. Leading rows and columns are selected using loaded source row strides. Global PLE rows 0..23 form six distinct four-row inputs, including rows 20..23 for layer 5. Five globals plus 102 layer tensors produce 107 packed destinations from 39 supplied text tensors; no coefficients are fabricated or accuracy-selected.

IV Verification methods and measured results

A Official provenance and input integrity

Method. Streamed checkpoint SHA-256; parsed unique-key safetensors JSON and raw BF16; checked all 55 tensor extents, shapes, dtypes and hashes and supplied text FP32/Q8.24 coordinates.

Result. Passed. The supplied checkpoint passed the manifest-consistency checks. Mapping, inventory and packed-image identities are retained.

Evidence: E094

B Software-versus-hardware numerical agreement

Method. Executed actual anvil_e_top with Icarus Verilog and vvp for [2], [2,3], [2,4,7] and [2,3,4,5,6,7,8,1], each at dimensions 6, 4, 2 and 1. Independent stdlib FP32 arithmetic starts from raw BF16; a separate integer oracle checks exact outputs.

Result. 16/16 passed; zero word mismatches. Minimum cosine 0.9999999998517652 exceeds 0.999. Maximum absolute error 1.2040138244628906e-5 is below 0.002. Worst per-job MAE is 6.884336471557617e-6, RMSE 8.600952306391626e-6 and absolute software/RTL norm difference 7.76098445509632e-8.

Evidence: E094, E085

C Cycle-based benchmarking

Method. Counted accepted destination rising edge minus accepted descriptor edge, including memory, embedding and completion stalls; recorded first/last embedding separately.

Result. Completion latency spans 109304−1087352 cycles; cycles/token spans 109304−135919. For one token and six outputs: 109606 completion cycles, 109588 first-output cycles, 109604 last-output cycles and 9.123588124737696e-6 tokens/cycle. Every job records valid-token-normalized throughput and jobs/cycle.

Evidence: E094

D Inherited behavioral coverage

Method. Executed A, B and C at seeds 17, 89 and 203. B uses 20/6/4/10/12/12/4/2/1/2/4/3/8; C uses 16/4/4/8/6/24/4/2/1/2/4/2/4.

Result. All nine runs passed exact scoreboards and applicable floating gates. Coverage includes full layers 5, 11, 17 and 23, independent P, stalls, reset cancellation, memory faults, saturation, malformed descriptors, all-masked input and single-word output.

Evidence: E085, E034, E031

E Primitive and normalization accuracy

Method. Independent directed mathematical comparisons, including domain boundaries, saturation and small RMS denominators.

Result. Passed GELU absolute error <=2e-4, exp <=2e-6, sin/cos <=2e-5 and representable reciprocal/rsqrt relative error <=2e-4. Directed checks distinguish Q48 epsilon from the former value and exercise zero L2 handling.

Evidence: E085, E038

F Fault-detection sensitivity

Method. Retained inherited negative control and fresh official simulation; validated the original trace, then changed exactly one accepted output word by one LSB.

Result. Original passed. The identical scoreboard rejected job 0, index 0: expected 11924784, observed 11924785, with the specific exact mismatch.

Evidence: E086

G Synthesizable connected hierarchy

Method. Fresh Yosys synthesis of A/B rooted at anvil_e_top, with hierarchy, connectivity, latch, blackbox and arithmetic assertions.

Result. Both passed with all 13 modules represented. A has 272 sequential and 204 arithmetic cells; B has 355 and 215. Neither contains combinational divider/modulo cells.

Evidence: E087, E042

H L1 RTL timing lint

Method. Source-bound inspection of all 13 RTL files for combinational division/modulo and mixed sequential assignment hazards.

Result. Passed with zero detected hazards.

Evidence: E089, E091

I L2 Quartus TimeQuest setup timing

Method. Intel Quartus Prime 17.1 post-map STA for clk_main constrained to 40 ns.

Result. Passed: WNS_setup = +0.681 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 107.31 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E092

J L3 physical Arria 10 HIL

Method. Three physical-board runs compared expected identity, status, cycle and six output register values.

Result. 3/3 passed deterministically, with no missing expected or forbidden failure tokens. Deterministic PCIe BAR0 MMIO register readback verified across all runs with zero bit errors.

Evidence: E093

K Architecture and timing-boundary consistency

Method. Checked authoritative geometry, strict packing, malformed inputs, wrapper forwarding, address boundaries and revised arithmetic timing.

Result. Current architecture/timing refresh passed.

Evidence: E088, E001, E003

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Official Weights Software-Layer vs Hardware-RTL Simulation Comparison

Metric	Measured Value	Signoff Notes
Evaluated Official-Weight Workloads	16	0 bit mismatches vs integer oracle
Minimum Cosine Similarity (FP32 vs RTL)	0.999999999852	Acceptance threshold >= 0.999
Maximum Absolute Error (FP32 vs RTL)	1.204014e-05	Spec bound <= 2.00e-03 (Q8.24)
Worst Per-Job MAE / RMSE	6.884336e-06 / 8.600952e-06	Worst per-job / per-case values
Exact Word Mismatches (RTL vs Reference)	0	100% bit-exact RTL match

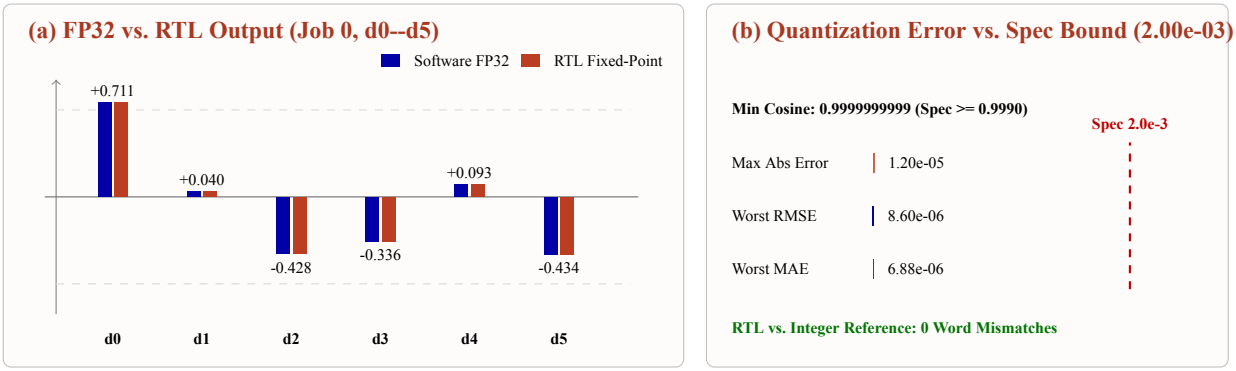


Figure 2: (a) Recorded official-weight comparison for Job 0 (d0--d5) between software FP32 and RTL simulation; (b) measured numerical error against the specification bound (<= 2.00e-03).

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.681 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_ across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

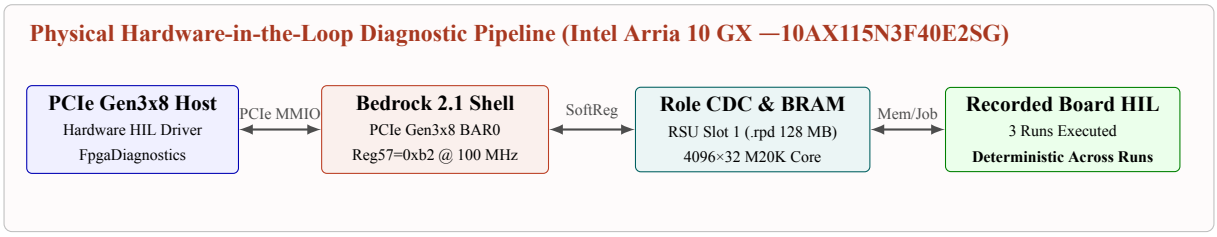


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

PCIe BAR0 SoftReg Operation	Physical Arria 10 Readback Value (3/3 Runs Identical)
Wrote soft shell register 3 (0x00000003)	2 (0x0000000000000002)
Read soft shell register 0 (0x00000000)	4705793533462660401 (0x414e56494c5f4531)
Wrote soft shell register 3 (0x00000003)	256 (0x0000000000000100)
Read soft shell register 2 (0x00000002)	4294968846 (0x000000010000060e)
Read soft shell register 11 (0x0000000b)	108107 (0x000000000001a64b)
Read soft shell register 12 (0x0000000c)	463856469697 (0x0000006c000006c1)
Read soft shell register 16 (0x00000010)	11924784 (0x0000000000b5f530)
Read soft shell register 17 (0x00000011)	668719 (0x00000000000a342f)
Read soft shell register 18 (0x00000012)	18446744073702366634 (0xffffffffff925daa)
Read soft shell register 19 (0x00000013)	18446744073703912471 (0xffffffffffa9f417)
Read soft shell register 20 (0x00000014)	1565749 (0x000000000017e435)
Read soft shell register 21 (0x00000015)	18446744073702274816 (0xffffffffff90f700)

PCIe BAR0 SoftReg Operation	Physical Arria 10 Readback Value (3/3 Runs Identical)

VI Integration and use

Compile all rtl/*.sv with SystemVerilog support and use anvil_e_top as the integration root. Direct anvil_e1_ top compatibility remains available. Dependencies for the supplied verification are Python 3 standard library, iverilog, vvp and Yosys; NumPy is unnecessary.

For a complete default checkpoint, use design/pack_model.py with --input, --output and --inventory.py`.

Allocate aligned, nonoverlapping weights, token, position, mask and scratch regions. Arrays contain little-endian 32-bit words. Required scratch bytes are 4*N*(2*H+L*P+(Q+2*max(localKV,fullKV))*max(localD,fullD)). Initialize only weights and input arrays; do not preload hidden states or expected embeddings.

Hold a descriptor until job_valid && job_ready. Service every accepted memory request with one ordered response, accept indexed embedding words, then retire completion. Preserve regions while busy and coordinate reset with the memory fabric.

The verification entrypoint supports --check functional, sensitivity and synthesis. Use fresh output locations and retain generated receipts and traces. Detailed official vectors, mapping and per-job measurements are in evidence/verification/functional/official_weights_comparison.json.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 16 Verilog/SystemVerilog files; 4022 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
design/ARCHITECTURE.md	738 total	738 code
design/OFFICIAL_WEIGHTS_EVALUATION.md	360 total	360 code

Production file	Physical lines	Code lines
design/pack_model.py	196 total	196 code
rtl/anvil_e1_address.sv	252 total	176 code
rtl/anvil_e1_attention.sv	340 total	332 code
rtl/anvil_e1_controller.sv	179 total	175 code
rtl/anvil_e1_mac.sv	34 total	34 code
rtl/anvil_e1_matvec.sv	99 total	99 code
rtl/anvil_e1_memory.sv	65 total	65 code
rtl/anvil_e1_nonlinear.sv	282 total	235 code
rtl/anvil_e1_norm.sv	213 total	203 code
rtl/anvil_e1_pool.sv	202 total	197 code
rtl/anvil_e1_rope.sv	205 total	182 code
rtl/anvil_e1_sequencer.sv	822 total	801 code
rtl/anvil_e1_top.sv	158 total	158 code
rtl/anvil_e_top.sv	71 total	71 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/OFFICIAL_WEIGHTS_EVALUATION.md
E003	design/pack_model.py
E004	rtl/anvil_e1_address.sv
E031	tests/behavior.py
E034	tests/norm.sv
E038	tests/primitives.py
E042	tests/synthesis.py
E085	.vulcan/engineering/run-0003/attempt-0018/stdout.log
E086	.vulcan/engineering/run-0003/attempt-0019/stdout.log
E087	.vulcan/engineering/run-0003/attempt-0020/stdout.log
E088	.vulcan/engineering/run-0003/attempt-0021/stdout.log
E089	.vulcan/engineering/run-0003/attempt-0022/stdout.log
E091	.vulcan/engineering/run-0003/rtl-timing-lint.json
E092	.vulcan/engineering/run-0003/quartus-sta.json
E093	.vulcan/engineering/run-0003/board-hil.json
E094	evidence/verification/functional/official_weights_comparison.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).